

DevOps lab Program 4: Integrate Git with Docker for Source-Controlled Application builds

```
1RV24MC075_Pragathi_h_r@dell: ~$ cd DevOps
1RV24MC075_Pragathi_h_r@dell: ~/DevOps$ cd prg4
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4$ npm init -y
Wrote to /home/tani/DevOps/prg4/package.json:

{
  "name": "prg4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "repository": {
    "type": "git",
    "url": "git+https://github.com/Pragathihr/prg4.git"
  },
  "bugs": {
    "url": "https://github.com/Pragathihr/prg4/issues"
  },
  "homepage": "https://github.com/Pragathihr/prg4#readme"
}

1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4$ nano index.js
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4$ npm install express

up to date, audited 68 packages in 1s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4$ nano Dockerfile
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4$
```

index.js

```
GNU nano 6.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration");
});
app.listen(PORT, () => {
  console.log("Server running at http://localhost:${PORT}");
});
```

Dockerfile:

```
11:54 es Terminal Nov 19 3:42 PM
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg4

GNU nano 6.2 Dockerfile
FROM node:18 AS build
WORKDIR /app
COPY package*.json
RUN npm install
COPY ..

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD["node", "index.js"]
```

Git commands:

```
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git init
Reinitialized existing Git repository in /home/tani/DevOps/prg4/.git/
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git add .
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git commit -m "initial commit"
[main 264e40a] initial commit
 2 files changed, 11 insertions(+), 2 deletions(-)
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git branch -M main
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git remote add origin https://github.com/Pragathihr/prg4.git
error: remote origin already exists.
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$ git push -u origin main
Username for 'https://github.com': Pragathihr
Password for 'https://Pragathihr@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Pragathihr/prg4.git/'
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg4$
```

Committed into Git:

